A Verbatim-Grounded, Field-Neutral Tag Layer for Cross-Disciplinary Reading of Academic Literature: A Proof of Format

Shir Sivroni

Department of Mathematics and Computer Science, The Open University of Israel · shirsivroni@gmail.com · Preprint, July 2026 · v8 audit revision

Abstract. Academic knowledge is written almost entirely in natural language, which makes it readable by humans but hard to query *across* disciplines that use different words for related things. We describe a tag layer that assigns every paper the same small set of **field-neutral facets** — claim type, phenomenon (a normalised join key), claim relations, replication status, and limitations — under one non-negotiable rule: **every tag is grounded in an exact, verbatim sentence from the paper's own text**. We report a proof-of-*format* on a curated corpus of 126 papers spanning 39 disciplines (1866–2023), of which 116 were fully faceted. The format is feasible and verifiable: 127 claim-relations and 87 limitations, each substring-checked against the source. It is also honest about what it did *not* do: a novelty scan surfaced no cross-disciplinary link that the literature had not already drawn, and the phenomenon facets, written as specific verbatim-anchored strings, rarely collide across fields — indicating that a **canonical join vocabulary**, not free text, is the missing ingredient for surfacing new links. We argue the value of the layer is as a **compass for "far reading"** that points where slow close reading may be worthwhile, not as an answer engine. Finally, we validate the fix the null result points to: a small controlled vocabulary of statistical terms supplies shared cross-field join keys, with the broadest audited term spanning **31 disciplines** (§5). A complete re-audit of 519 legacy paper–term assignments rejected 99 (19.1%) and retained 420 semantically accepted assignments; because source availability and exact excerpt grounding are incomplete, the primary counts and a stricter exact-local-body sensitivity analysis are reported separately. As a first structural test, 16 papers across 8 disciplines already group under one canonical phenomenon (priming) despite each field naming it differently. A grounded audit of that cluster then asks the harder question — does each field's priming actually carry the definition's three features? — and finds **association** and **modulation** verbatim-grounded in 16/16 papers and **secondariness** in 14/16, the two exceptions being genuine boundary findings (§5.1, Figs. 5–6).

1. Motivation

Most scholarly literature is natural-language prose. Cross-disciplinary work is therefore bottlenecked not by a lack of relevant results elsewhere, but by the difficulty of *finding* them: the same mechanism, method, or phenomenon appears under different names in different fields. Existing structured resources — ontologies, citation graphs, keyword indexes — either require heavy per-field curation or capture topic words that are, by construction, discipline-specific and so useless as join keys.

2. The idea

We tag each paper with a fixed, deliberately small set of **facets chosen to be field-neutral** (they describe the *shape* of a contribution, not its topic): `claim-type` (existence / mechanism / method / prediction / critique / null-result), `phenomenon` (a normalised name intended as a cross-field join key), `claim-relations` (builds-on / extends / contradicts / replicates, each toward a named target), `replication-status`, and `limitations`. The defining constraint is **verbatim grounding**: a tag is an interpretation of natural language, so the design requires every non-trivial tag to carry a licensing excerpt and a resolvable provenance state. The present audit reports where the legacy data meets that requirement and where it does not. This is what separates the intended layer from an ungrounded ontology (whose assertions cannot be checked against a source) and from a dense embedding (which is not human-auditable). The layer serves humans (HTML views with the licensing sentence attached) and machines (a single JSON record per paper) at once.

2.1 The data structure: one property graph over a grounded tag layer

The layer can be represented as a **property graph**, built in three layers (Fig. 1). At the base is the **corpus** — 126 papers as raw, natural-language text, one record each. Over it sits the **grounded tag layer**: every paper carries the same field-neutral facets, and each non-trivial tag stores its licensing excerpt and provenance status. The implemented graph export reads these records as a graph — **nodes** are papers, tags, disciplines, and canonical phenomena; **edges** are `HAS_TAG` (carrying its evidence), `IN` (paper→discipline), `ALIAS_OF` (a field's local name→a canonical phenomenon), `CONGRUENT_TO` (a scored 0–1 correspondence between two phenomena), and projected cross-field links. The current export implements `IN`, `HAS_CLAIM`, `HAS_PHENOMENON`, and audited `HAS_STATISTIC` edges; `ALIAS_OF`, `CONGRUENT_TO`, and projected links remain a proposed extension, not a measured output.

Two properties support scaling. First, verification becomes a **set query** rather than manual reading: “does each priming phenomenon carry association, secondariness and modulation?” is the intersection of four tag-sets, and its *complement* lists the exceptions automatically (§5.1). Second, the implementation maps each facet value to a sorted postings list. Dictionary lookup is expected O(1), while an ALL query intersects lists in $\Theta(\Sigma|P_i|)$ time plus output work; a common tag, unfiltered listing, or complement can therefore still be $\Theta(N)$. Selective queries scale by avoiding unrelated records, not by becoming independent of corpus size. Standard graph algorithms may run over the exported adjacency list with their ordinary graph-dependent costs (for example, $O(|V|+|E|)$ for a full traversal). The same query syntax can be used on thousands of papers, but runtime depends on postings lengths and graph size (§5.1, Fig. 6).

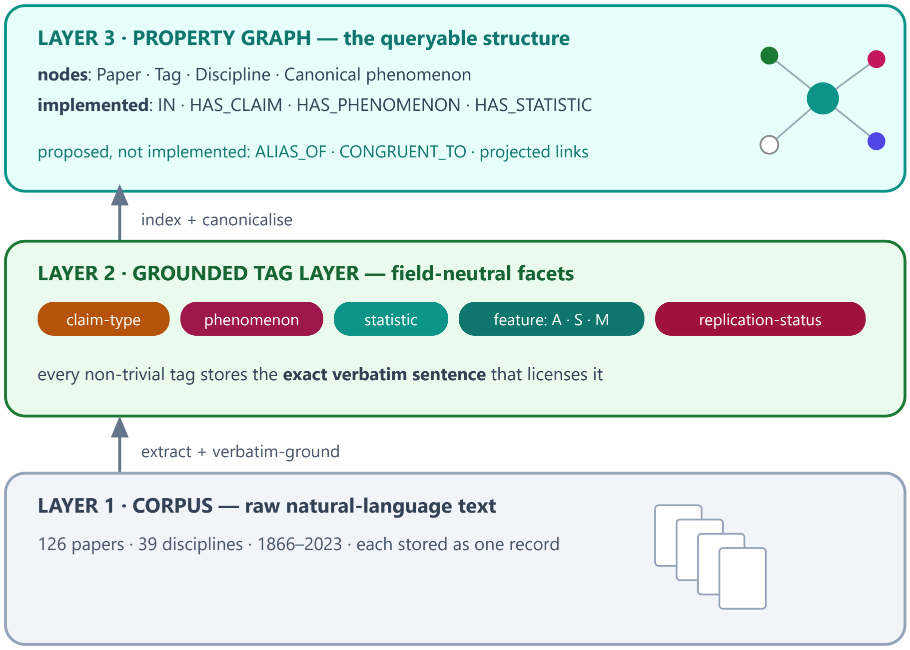


Fig. 1. The data structure: one **property graph** in three layers. A corpus of raw text (Layer 1) carries a grounded tag layer (Layer 2), which is read as a graph of nodes and edges (Layer 3) over which verification becomes a query. Full specification in the project's `GRAPH_SCHEMA`.

Concretely (Fig. 2), each paper is one **record** — a JSON document holding its facets and, for each audited statistical tag, the evidence excerpt, decision, reason, role, and provenance hashes. From these records the reproducible build derives two classic structures: an **inverted index** (a dictionary from a facet value to a sorted postings list of numeric document identifiers) and a typed **adjacency list** containing only relations present in the records. Multi-tag search is a sorted-list intersection or union; graph analyses can consume the adjacency export.

① one paper = one record (document store — a dict)

```
{
  "id": "newman2003-complex-networks",
  "discipline": "network-science", "year": 2003, "license": "arxiv-open",
  "title": "The Structure and Function of Complex Networks",
  "phenomenon": "complex networks (shared structure of real networks)",
  "content_tags": {
    "statistics": [{ "term":"correlation", "symbol":"r",
        "evidence":"For the data of Table III we find r = 0.621." }],
    "key_terms":  ["networks","degree","clustering","vertices","edges"] } }
```

② inverted index (facet → sorted postings) — expected O(1) key lookup

```
"statistic:correlation"      → [ 7, 12, 18, ... ] (39 papers · 22 disciplines)
"claim:method"               → [ 1, 3, 6, ... ]                     (55 papers)
"discipline:network-science" → [ 83, 84 ]                           (2 papers)
ALL = intersect; ANY = union; complement = universe difference
merge cost = Θ(Σ|P_i|) + output; worst case Θ(N)
```

③ implemented adjacency export (node → typed neighbours)

```
paper:newman2003 → [ IN:discipline:network-science,
                         HAS_STATISTIC:statistic:correlation, ... ]
not exported: ALIAS_OF · CONGRUENT_TO · projected links
```

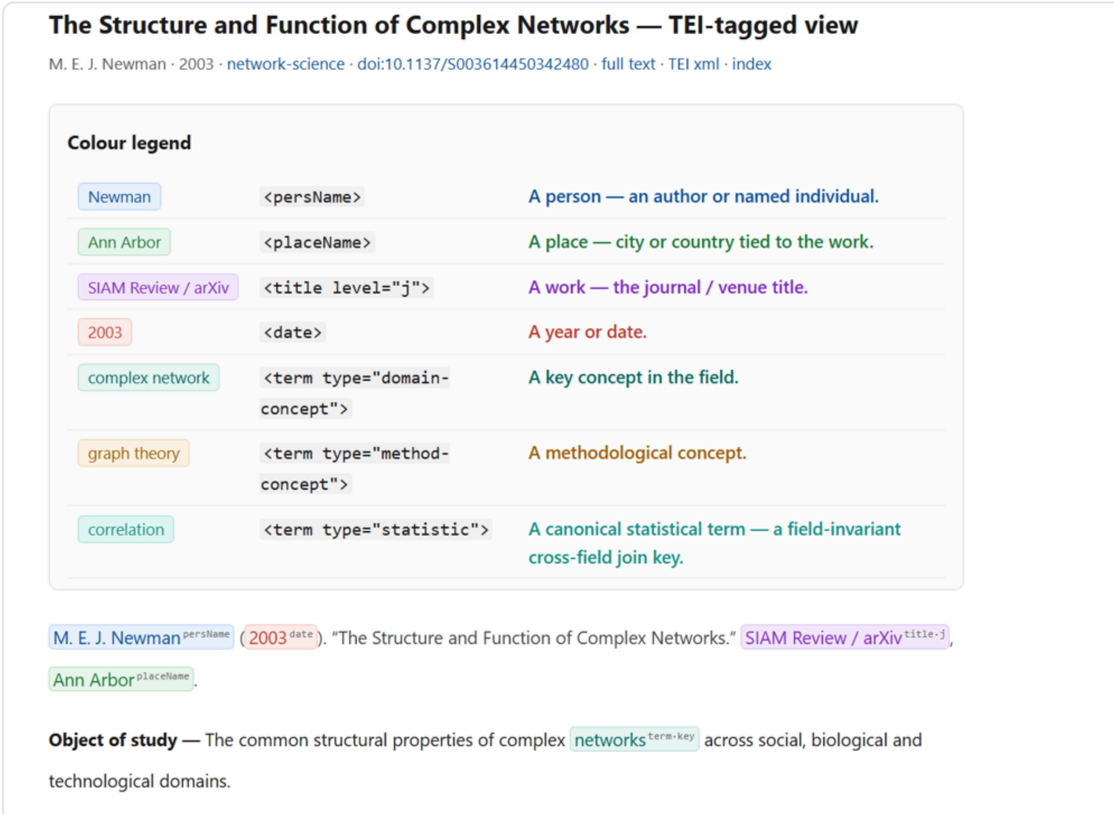
Fig. 2. Implemented data structures. Each paper is one canonical JSON record. The build derives sorted postings and lazy metadata shards for search, plus a typed adjacency export. Dictionary lookup is expected O(1), but merging postings depends on their lengths and can be $\Theta(N)$; no corpus-size-independent complexity is claimed.

3. Corpus and method

The corpus is a curated, deliberately broad sample — an *existence proof of the format and procedure, not a claim of completeness or representativeness*. Each paper is stored as a single canonical record; the implemented build audits and validates statistical annotations, recalculates the reported quantities, derives search shards and postings, exports a typed adjacency list, and regenerates the audit and paper views. The original faceting process intended character-level grounding, but the present re-audit shows that this invariant was not consistently preserved. Grounding status and hashes are now explicit fields, and unverified excerpts are reported rather than silently treated as exact.

Data quality. Of 126 papers, **10 (7.9%)** could not be faceted — all because of a *garbled source text layer* (image-only scans, dropped characters, interleaved sources), not because their content resisted the method. On clean text the method faceted 116/116. This failure rate therefore measures digitisation/sourcing quality, not the tagging method; we record such papers under an explicit `untaggable` tag rather than dropping them silently.

How a record is read back. Every open-access paper is rendered as a **colour-tagged reading view** (Fig. 3): the same facets shown inline over the paper's own text, each tag colour-coded and labelled with its type, so a human can audit every tag against the sentence that licenses it. Paywalled papers are linked by DOI, not reproduced.


The Structure and Function of Complex Networks — TEI-tagged view

M. E. J. Newman · 2003 · network-science · doi:10.1137/S003614450342480 · full text · TEI xml · index

Colour legend

Newman	`<persName>`	A person — an author or named individual.
Ann Arbor	`<placeName>`	A place — city or country tied to the work.
SIAM Review / arXiv	`<title level="j">`	A work — the journal / venue title.
2003	`<date>`	A year or date.
complex network	`<term type="domain-concept">`	A key concept in the field.
graph theory	`<term type="method-concept">`	A methodological concept.
correlation	`<term type="statistic">`	A canonical statistical term — a field-invariant cross-field join key.

M. E. J. Newman persName (2003 date). "The Structure and Function of Complex Networks." SIAM Review / arXiv title·j, Ann Arbor placeName.

Object of study — The common structural properties of complex networks term·key across social, biological and technological domains.

Fig. 3. The colour-tagged reading view (Newman 2003, an open-access paper): a colour legend plus the paper's own text with inline, type-labelled tags (persName, date, title, key-term, statistic…), each auditable against its sentence. **Click the figure** for the live page: shir-openu.github.io/meta-tagging-showcase/papers/newman2003….

4. Results

116 papers across 38 disciplines were fully faceted.

claim-type	method 55 · existence 42 · mechanism 9 · critique 8 · prediction 2 · null-result 0
replication-status	untested 113 · replicated 3 · (failed/contested 0 — see §7)
grounded relations	127 claim-relations, 87 limitations — all substring-verified

The central honest finding. As free-written strings, the 113 distinct `phenomenon` values almost never coincide across fields: only **1** pair of disciplines shares a phenomenon string verbatim. The bridging one might hope for (e.g. a "priming" or "power-law" motif recurring across many fields) is real at the level of shared *concepts*, but the strict, verbatim-anchored phenomenon facet is worded too specifically to collide. The implication is concrete: the join key needs a **canonical vocabulary** (a controlled, normalised phenomenon namespace) layered over the grounded free text — not more papers. A separate novelty scan, checking every cross-field candidate against the existing literature, found **no** connection that had not already been drawn; we report this null result plainly. The fix this points to — a canonical vocabulary — is tested directly in §5.

5. Audited statistical terms as cross-field join keys

§4 argued that the missing ingredient is a *canonical vocabulary*, not merely more papers. We therefore tested a frozen 24-term `statistic` vocabulary (`p-value`, `standard-deviation`, `confidence-interval`, `effect-size`, `correlation`, `regression`, `hypothesis-testing`, …). Statistical terms usually have more stable cross-field semantics than topic phrases, but they are not perfectly field-invariant: estimators, tests, parameterisations, and reporting conventions vary. A shared term is therefore a retrieval key, not evidence that two papers used an identical operation or reached a substantively comparable result.

Audit protocol. The unit was a paper–term assignment. A term was semantically accepted only when the supplied evidence or locally available article body supported an unambiguous statistical or probabilistic sense. Generated statistics summaries, bibliography-only matches, lexical collisions, non-inferential uses of “significant”, class probabilities labelled p, and symbols such as H_0 without an explicit null-hypothesis sense were excluded. The audit resolved all **519** legacy assignments: **420 accepted** and **99 rejected** (19.1%), with no unresolved audit decisions. The tag indicates an accepted occurrence, which may be a focal use or a substantive mention; it does *not* by itself assert that the paper employed that method.

Grounding sensitivity analysis. The legacy corpus did not preserve equally strong provenance for every excerpt. Of the 420 semantically accepted assignments, **272** have an exact normalized excerpt located in the locally stored article body; 97 belong to papers whose source text is not stored locally, 41 excerpts were not located exactly in the extracted body, and 10 supplied excerpts resolve to a reference block even though a separate body occurrence supported the semantic decision. Consequently, the full accepted analysis and the stricter exact-local-body subset are reported side by side. The latter is a provenance check, not an unbiased lower bound, because local source availability is non-random.

Across the full accepted set, at least one statistical key occurs in **100 of 126 papers** and **37 of 39 disciplines**. The strict exact-local-body subset contains **272 assignments in 77 papers across 32 disciplines**. The recalculated term-level results are:

join key	accepted: papers / disciplines	exact local body: papers / disciplines
`mean`	59 / 31	39 / 25
`standard-deviation`	34 / 24	23 / 15
`correlation`	39 / 22	23 / 15
`regression`	33 / 19	24 / 16
`statistical-significance`	27 / 19	18 / 13
`p-value`	24 / 17	10 / 9

The previously reported `p-value` + `standard-deviation` neighbourhood also changes: it contains **10 papers across 10 disciplines** in the accepted analysis, not 14 papers across 13 disciplines. Requiring both excerpts to be exact local-body matches leaves **3 papers across 3 disciplines**. These co-occurrences define candidates for comparison; they do not establish substantive equivalence.

The revised result is narrower than the original claim but still supports the structural point: a controlled namespace produces more cross-field joins than highly specific free-text phenomenon strings. It is evidence for the retrieval design, not proof that statistical vocabulary is field-invariant or that the joins are scientifically novel. The natural next step is a similarly audited canonical `phenomenon` namespace.

A first structural test already supports this. Our priming schema — the association / secondariness / modulation slots — groups **16 papers across 8 disciplines** (neuroscience, cognitive and social psychology, behavioral economics, economics, marketing, education, linguistics) under one canonical phenomenon, even though each field names it differently. Detected on the tag graph, it is an existence proof that a canonical phenomenon layer, grounded in shared *structure* rather than shared *words*, connects fields the free-text facet left disjoint (Fig. 4).

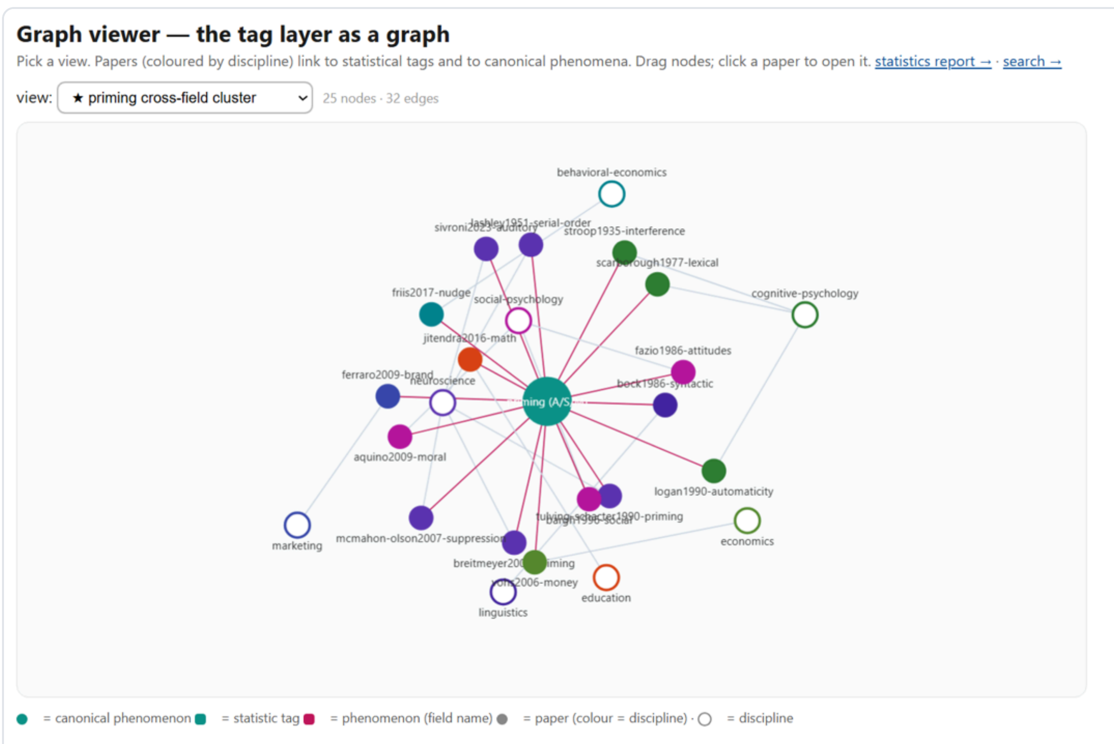


Fig. 4. The priming cross-field cluster in the interactive graph viewer (**real data**): one canonical *priming (A·S·M)* node at the centre links to 16 papers, coloured by discipline, wired out to their eight fields (neuroscience, cognitive and social psychology, economics, behavioral economics, marketing, education, linguistics) — though each field names it differently (interference, automaticity, nudge, money-priming, syntactic priming). 25 nodes · 32 edges. **Click the figure** for the live interactive graph: shir-openu.github.io/meta-tagging-showcase/GRAPH.html (opens locally as `GRAPH.html`).

We are careful about the claim: a shared *method* vocabulary is a weaker signal than a shared substantive finding — it marks where a connection *may* lie, consistent with the compass framing (§6), not a discovery.

5.1 A grounded test: does each field's priming actually carry the three features?

Showing that "priming recurs across many fields" is trivial and long known; it needs no tool. The non-trivial, *falsifiable* question is whether *each* field's priming phenomenon actually instantiates the three defining features of the meta-priming definition — **A · association** (a prime–target relation), **S · secondariness** (the prime is incidental / prior / task-irrelevant, not the reported object itself), and **M · modulation** (the outcome differs from a baseline) — *grounded verbatim in that paper's own text*. We turned each feature into a checkable, verbatim-anchored tag and audited all 16 priming papers across the 8 disciplines.

Association and modulation are *structural*: they are read directly from the tagged prime→target and baseline→outcome slots, and hold in **16/16** papers. Secondariness is the *discriminating* test — the prime must be textually marked as incidental, prior, or task-irrelevant, and it **can fail**. It is grounded in **14/16** papers; the two exceptions are findings, not noise:

- **Education** (schema-based mathematics instruction): the "prime" is a *deliberately taught* schema — the object of instruction, not an incidental cue — so secondariness is weak by construction.
- **Neuroscience** (serial-order motor control): motor *pre-activation* is a constitutive part of the produced action, not a task-irrelevant prime.

This is the payoff of grounding: the layer does not rubber-stamp the definition, it **locates the boundary** where a field's usage of "priming" stretches it. The verification is a tag intersection — `phenomenon = priming AND feature:association AND feature:secondariness AND feature:modulation` — whose *complement* surfaces the exceptions automatically. In the implemented sorted-postings index, key lookup is expected O(1), intersection costs $\Theta(\Sigma|P_i|)$ plus output, and the universe difference needed for a complement can cost $\Theta(N)$. The query remains practical for thousands of papers when the selected postings are sparse, but it is not independent of corpus size (Fig. 6).

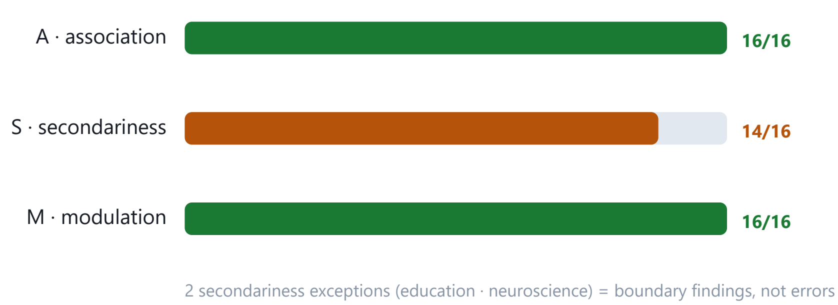


Fig. 5. Grounded A·S·M verification of the 16-paper priming cluster. Association and modulation are read from the structural slots (16/16); secondariness is a textual test that can fail (14/16). **Live interactive version** (per-paper evidence, drill-down): shir-openu.github.io/meta-tagging-showcase/PRIMING_SCALE.html.

What it looks like at scale. The same indexed query can be evaluated over a thousands-paper corpus, with cost determined by the selected postings and output: one canonical *priming* node wired to its three defining features, surrounded by the disciplines that study it, each field coloured by whether all three features hold (green) or one is under-met (amber — an exception to investigate). The corpus itself is one graph — papers, tags, disciplines, and canonical phenomena as nodes; grounded tags and cross-field projections as edges — over which such questions become queries rather than manual reading.

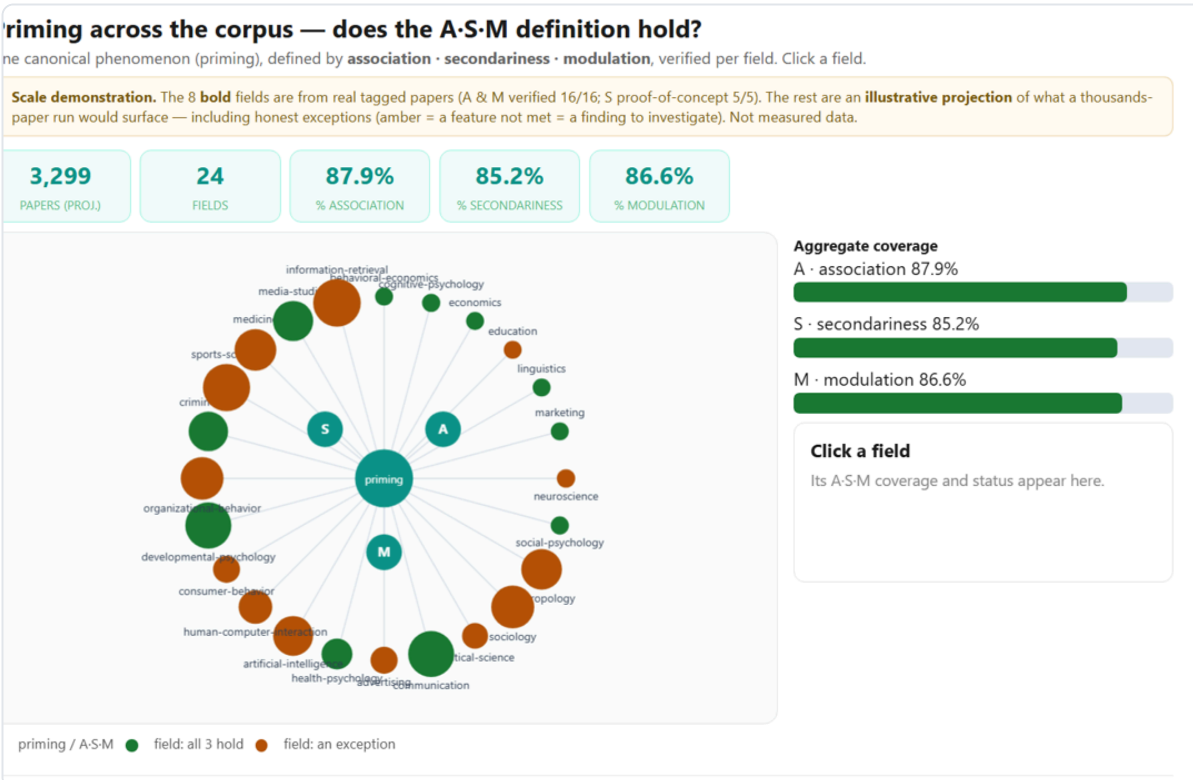


Fig. 6. The same A·S·M verification as an **illustrative projection** to a thousands-paper corpus (24 fields): one canonical *priming* node wired to its three features, each field green if all three hold or amber if one is under-met (an exception to investigate). The eight bold fields carry the real audit numbers; the rest illustrate what a large run would surface — *not measured data*. **Click the figure** for the live interactive version: shir-openu.github.io/meta-tagging-showcase/PRIMING_SCALE.html.

5.2 Querying the layer: an implemented static, client-side search

The revised search ships as static assets served over HTTP: a small manifest, a postings dictionary, and lazy-loadable metadata shards (500 records per shard by default). It needs no application server or query-time database, but it is no longer described as one embedded self-contained file. The interface exposes exactly the facets the layer defines. **Input:** the reader composes a query from free text (title / author / key term), the statistics join-keys grouped by family (central tendency, spread, association, inference, advanced), claim-type, and discipline, and chooses `ALL` (intersection) or `ANY` (union) for the selected statistical keys. **Output:** every paper matching the query, *across all fields at once* — each result card shows the paper's discipline, claim-type, the matched tags, its key terms, and a link to the colour-tagged full text with the evidence excerpt (Fig. 7). For one selected facet, dictionary lookup is expected O(1); ALL and ANY queries merge sorted postings in $\Theta(\Sigma|P_i|)$ time plus output work. An unfiltered listing or a common facet can be $\Theta(N)$, and a complement requires a universe difference that can also be $\Theta(N)$. Record shards are fetched only for the current result page. Thus the implemented gain is reduced scanning and transfer for selective queries, not a size-independent asymptotic bound.

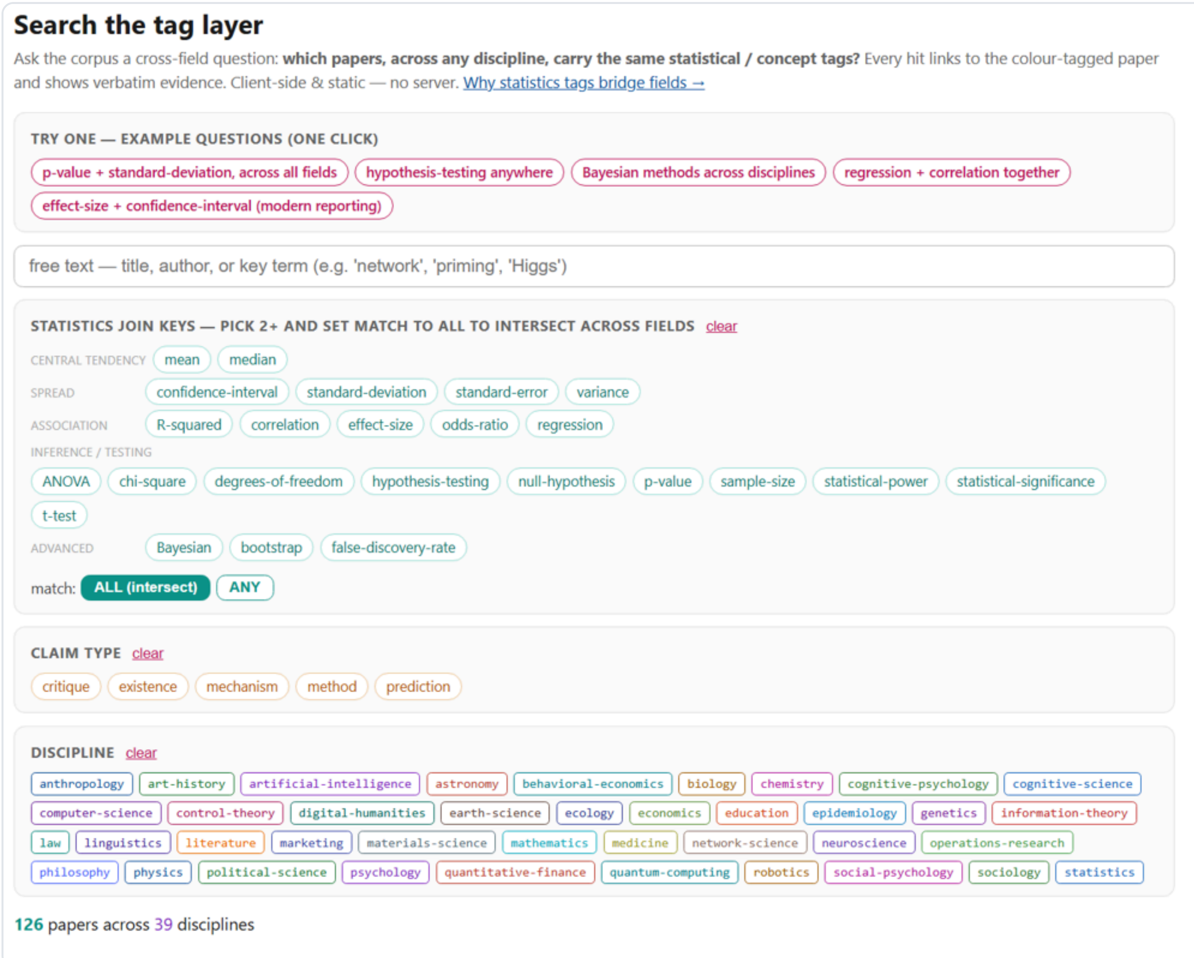


Fig. 7. The static, client-side search. The reader intersects field-neutral tags — statistics join-keys (grouped), claim-type, discipline, free text — with an `ALL` / `ANY` switch; results are papers from any field, each with its matched tags, key terms, and a link to the colour-tagged evidence. No application server; static HTTP hosting is required. **Click the figure** for the live search: shir-openu.github.io/meta-tagging-showcase/SEARCH.html.

6. Framing: a compass, not an answer

The layer is intended to enable *far reading* — a corpus-wide overview in the spirit of distant reading [1] — that indicates *where* a cross-disciplinary connection may lie, so that a scholar knows where to invest the slow *close reading* that can actually establish it. A surfaced link is a hypothesis to investigate, never a claimed discovery; because one cannot prove the absence of a prior connection, every candidate is presented with its literature check, not asserted.

7. Limitations

- The `statistic` facet (§5) records an accepted statistical or probabilistic occurrence, not necessarily a method the paper used. Sharing a key marks a candidate neighbourhood, a much weaker signal than a shared substantive result. The full audit rejected 99 of 519 legacy assignments; semantic judgment error may remain.

- Evidence provenance is incomplete. Only 272 of 420 accepted statistical assignments currently have an exact normalized excerpt located in a locally stored article body. The other provenance states are reported separately and must not be described as exact-body verified.
- Tagging currently depends on a large language model to read prose; the grounding guard bounds but does not eliminate interpretation error.
- The corpus is curated for breadth, not sampled representatively; distributions above describe *this* corpus only.
- `replication-status` and `limitations` are grounded in each paper's *own* text; external replication outcomes (e.g. well-known failed replications) are deliberately *not* injected, because they are not grounded in the source — a candidate future extension, not a silent edit.
- The phenomenon join key needs canonicalisation before it can surface cross-field links at scale (§4).
- The implemented graph export does not include `ALIAS_OF`, `CONGRUENT_TO`, projected links, or graph-analysis results. The priming projection is a separate curated analysis and Fig. 6 is explicitly illustrative.
- Full-text HTML is unsuitable as the primary Git-backed store for a much larger corpus. Production deployment should keep compact, versioned metadata and build code in version control, put redistributable full text in content-addressed object storage, and serve postings and immutable record shards through static hosting. The current in-memory build is intended for thousands to low tens of thousands of records, not unbounded scale.
- Novelty claims are unfalsifiable in the strict sense; we can search and report, not prove a negative.

8. Relation to prior work

The layer sits between text-encoding standards (TEI), stand-off annotation models (the W3C Web Annotation Data Model [2]), machine-readable scientific assertions (nanopublications [3]), scholarly knowledge graphs, and computational distant reading [1]. Its distinguishing commitments are the union of (i) mandatory verbatim grounding of every tag, (ii) facets chosen to be field-neutral join keys rather than topic descriptors, and (iii) a hypothesis-testing stance about whether such a layer can surface undrawn cross-field links — reported here honestly: free-text `phenomenon` strings do *not yet* collide across fields, while an audited controlled vocabulary of statistical terms (§5) creates cross-field retrieval joins. These joins remain semantically and provenance-qualified, pointing the way to canonicalising the rest without claiming field invariance.

9. Availability

A public showcase provides the report and the full colour-tagged text for open-access papers, with paywalled originals linked by DOI (not reproduced). The revised repository adds a canonical corpus JSON, an explicit audit-decision file, schema validation, complete Section 5 recalculation, sorted postings, lazy metadata shards, a typed adjacency export, and one reproducible build command. It includes a **static, client-side search** over the tag layer and a generated statistics audit report. Live showcase: shir-openu.github.io/meta-tagging-showcase/. Archived on Zenodo — DOI: 10.5281/zenodo.21429867.

Acknowledgements. This work grew out of the *Beyond Words* Digital-Humanities-and-Social-Sciences (DHSS) summer school — a programme devoted throughout to the **tagging and encoding of natural language**: close reading, TEI/XML text-encoding, and stand-off annotation. I thank its organisers, instructors, and participants, whose exercises and discussions seeded the idea of a single, uniform tag layer. This work was carried out independently, without research funding.

1. F. Moretti, *Distant Reading*, 2013 — the "distant reading" framing this work adapts as "far reading."
2. W3C, *Web Annotation Data Model*, Recommendation, 2017 — stand-off annotation with explicit targets.
3. P. Groth, A. Gibson, J. Velterop, "The anatomy of a nanopublication," 2010 — minimal machine-readable assertions.

Preprint · v8 audit revision · July 2026 · licensed CC BY 4.0 · DOI: 10.5281/zenodo.21429867 · cite via CITATION.cff in the project repository. v7 re-audits all 519 legacy statistical assignments, recalculates §5, separates semantic acceptance from exact-local-body grounding, implements a sharded sorted-postings search and reproducible build pipeline, exports only supported graph relations, and corrects the architecture and complexity claims. Earlier versions introduced the priming A·S·M analysis and the statistical join-key experiment.